> *Laravel singletons persist **only for the duration of a single request lifecycle**. In development mode (via `php artisan serve`), the container is reloaded on each request, so singleton behavior is not preserved across refreshes. To observe persistence across requests, consider using **Laravel Octane** or test by resolving the singleton multiple times within the same request.*

5.12 Exercise 8: Contextual Binding with Dependency Injection in Laravel 12

5.12.1 Description

In this lab, we'll create a scenario where different controllers require different implementations of the same interface. We'll use **contextual binding** to tell Laravel's **Service Container** which class to inject based on the controller.

5.12.2 Objectives

By the end of this lab, you will:

- Understand and implement **contextual dependency injection**
- Create an interface with multiple implementations
- Use `when()->needs()->give()` in a service provider
- Render injected results in a simple Blade view

5.12.3 Prerequisites

Ensure the following are installed:

- Laravel 12
- PHP ≥ 8.2
- Visual Studio Code
- Basic Laravel knowledge (routes, controllers, Blade)

> *This lab uses **PHP 8.4**, **Laravel 12** and **Visual Studio Code** as the primary editor.*